Credit Risk Prediction Using Machine Learning

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Abstract

In this project, we will build credit risk prediction models using machine learning and evaluate their accuracy. We will use Logistic Regression and Decision Tree to predict whether applicants belong to good or bad credit-risk categories. After that, we will compare the performance of both models and determine whether the results meet our expectations. Lastly, we will explore the ethical concerns of using machine learning for credit risk prediction.

1 Introduction and Problem Statement

Predicting credit risk is important for financial institutions because it helps them determine the likelihood that applicants will repay their loans. Using the German Credit dataset, we will determine whether machine learning can accurately predict the credit risk of loan applicants. We will compare two machine learning algorithms to determine which one performs better and whether their results are reliable. We will also discuss the ethical concerns of using machine learning for credit risk prediction, including potential bias and incorrect predictions.

2 Dataset and Methodology

2.1 Dataset

The team will use the Statlog German Credit dataset available in the UCI Machine Learning Repository (1). The dataset contains 1,000 records, 20 features, and two credit-risk categories: good and bad. Some of the features include loan amount, loan duration, credit history, and employment information.

2.2 Data Preprocessing

To preprocess the dataset, categorical variables will be encoded, and numerical features will be scaled when appropriate. Furthermore, the dataset will be divided into training, validation, and testing sets using an 80/10/10 split, respectively.

2.3 Tools

The team will use Python and scikit-learn (2) to implement the following machine learning algorithms:

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- Logistic Regression: Used to predict the probability that an applicant belongs to a specific credit-risk category.
- Decision Tree: Used to classify applicants by learning decision rules based on their characteristics.

For a fair comparison, both models will be trained and evaluated using the same data splits. Lastly, the team will use Streamlit to develop an interactive application where users can enter applicant information and receive a predicted credit-risk category.

3 Evaluation and Expected Results

We will compare Logistic Regression and Decision Tree using accuracy, precision, recall, F1 score, and confusion matrices, treating bad credit risk as the positive class. Both models will use the same stratified 80/10/10 training, validation, and testing split. Preprocessing will be fitted on training data, validation data will guide tuning, and testing data will measure final performance. Results will also be compared with a majority class baseline. We expect both models to identify useful credit risk patterns, although their relative performance will be determined experimentally. We will also discuss common errors and signs of overfitting.

4 Ethical Consideration and Expected Deliverables

4.1 Fairness, Bias, and Privacy

Using machine learning to predict credit risk can be helpful, but there are also some concerns we need to consider. One important concern is fairness, because the models learn from existing data, which may contain patterns that lead to unfair predictions for certain groups of people. We will look at possible bias in the results and discuss how incorrect predictions could affect someone applying for a loan. Privacy is another concern, since credit information can be sensitive. We will also explain why these models should be used to assist people in making lending decisions rather than making decisions without human involvement.

4.2 Ethics in Problem Statement

Ethically, the problem statement itself should have a reason for pause, because the its output can influence wether a real person obtains a loan. The project uses a decision tree and logistic regression classifier which denotes a class of A (predicted default) or B (predicted repayment). We do not prescribe that a given sample should or should not be approved or denied for a loan. Lenders have an interest in managing loss from loans, which legitimizes the use-case of our project. However, this model should be used as an *input* rather than a final determination. The US federal law reflects the need for accountability in a lender's choice. For example, in the Equal Credit Opportunity Act (3), which requires lenders to state specific reasons for denying an application. The models we chose to use in our paper reflect this need for transparency, due to the ability for a human to retrace the decision making process.

4.3 Deliverables

By the end of this project, we plan to build two machine learning models, Logistic Regression and Decision Tree, using the German Credit dataset. We will compare their performance using accuracy, precision, recall, and confusion matrices. We also plan to create a simple Streamlit application where users can enter applicant details and see the predicted credit risk. Our final report will include the results, comparisons between the two models, and any limitations we notice. We will also prepare a presentation to explain our work and what we learned from the project.

References

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